Rohit Kumar Dubbaka

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Research Interests

Analog & Mixed-Signal IC Design · Bio-Potential Analog Front-Ends (AFEs) · Low-Noise Sensor Interfaces · Neuromorphic & Biomedical Circuits · Hardware-Software Co-Design

Education

Amrita Vishwa Vidyapeetham — Coimbatore, India

B.Tech, Electronics & Communication Engineering | CGPA: 7.64 / 10 | First Class — *Aug. 2022 – May 2026*

- *Relevant Coursework:* VLSI Design, Analog Electronic Circuits, Linear Integrated Circuits, Mixed-Signal Design, Microprocessors & Interfacing, Embedded Systems.
- *Strong Lab Performance:* Analog Electronic Circuits Lab (A), VLSI Design Lab (A).

Publications & Working Papers

Manuscript in Preparation — [Draft] — 2026

- [1] R. K. Dubbaka, A. Isukapalli, S. Harshath, and L. Vidhya, "Hands-Free sEMG-Based Speech-Controlled Wheelchair," in preparation, Amrita Vishwa Vidyapeetham, 2026.
  Full draft, schematics, and raw bench validation dataset available via link.

Professional Experience

Delta IoT Solutions Pvt. Ltd. — Hyderabad, India

Embedded Systems Engineer (Full-Time) — *Oct. 2026 – Present*

- Managing PCB assembly quality across a 235-unit IoT customer batch; authored BOMs, cost models, and assembly/build documentation adopted as team standard.
- Diagnosed and restored a Charmhigh SMT pick-and-place machine and a 3D printer, bringing in-house assembly and prototyping capability back online.
- Handled the design update and procurement for a new version of the in-house IO-expander design.

Delta IoT Solutions Pvt. Ltd. — Hyderabad, India

Embedded Hardware Intern — *Feb. 2026 – Aug. 2026*

- Engineered a pluggable 4-layer LTE/GPS module (55×44 mm) in KiCad with 50 Ω controlled-impedance RF layout, DFM/DFA review, and bench validation; achieved full LTE signal strength confirmed on bench.
- Developed an ultra-compact hot-swappable power board 21×24 mm sustaining 2.5 A transient bursts for a micro 2.3×1.4 mm DSBGA/WLCSP buck converter IC; architecture integrated into company's upcoming IoT board.
- Reconstructed complete schematic of a competitor's industrial IoT device via DMM and component datasheets, identifying power topology inefficiencies and directly driving a strategic upgrade to the company product BOM; performed in-field device deployment and firmware updates.
- Built breakout and test boards for LTE module and BGA/QFN devices; sourced, assembled, and revived prototype boards for bench bring-up and fault isolation.

Research & Academic Projects

Hands-Free sEMG-Based Speech-Controlled Wheelchair | [Link] Jun. 2025 – Mar. 2026

- *B.Tech Thesis* | Advisor: Ms. L. Vidhya.
- Developed a 3-stage discrete AFE (3-op-amp INA → 6th-order Bessel BPF 20–300 Hz → 50 Hz KRC notch) for laryngeal sEMG bio-potentials during silent-speech. Bench-validated gain: ∼70 dB, CMRR ∼90 dB, and noise floor via DSO/FFT; wavelet denoising lifted SNR from 2.43 dB → 10.63 dB.
- Laid out a compact 2-layer PCB (60×25 mm) in KiCad with SMD 0805 passives.
- Collected a 40-subject / 6-command in-house dataset (14,598 silent-speech samples × 32 time/frequency features); ANOVA-selected features fed a tuned Random Forest (BayesSearchCV) achieving 88% test accuracy (F1 = 0.88).
- Implemented real-time streaming pipeline: PC DAQ/ML streamed via USART3 (115.2 kbps) to STM32-H753ZI (Cortex-M7, 480 MHz) running bare-metal C firmware for differential PWM motor drive (dual BTS7960 43 A H-bridges), 500 ms watchdog, and microsecond DWT cycle-counter ultrasonic safety interrupts (<50 cm); prototyped on-device C decision trees.

Deterministic Priority Arbiter for Multi-Channel Bus | [Link] Dec. 2024 – Mar. 2025

- Architected a 4-channel bus priority arbiter in VHDL combining a 4-to-2 priority encoder, a 4-bit cyclic barrel shifter, and a structural 4-stage D-FF ring counter with parallel loading to achieve deterministic, starvation-free token circulation.
- Synthesized on Artix-7 XC7A35T; verified single-cycle deterministic grant latency and 7.96 ns data path delay; occupied minimal footprint of 8 Slice LUTs (0.04%), 5 slices, and 8 registers across 14 user I/O pins.

Hackathons and Competitions

Aeromesh: Real-Time Edge Turbulence Classifier | [Link] Aug. 2026

- *CraftifAI Buildathon – Organizer Spotlight.*
  Trained and deployed an in-situ TinyML motion classifier for sub-10ms flight turbulence on ESP32 from live inertial data from the 6-axis IMU - MPU6050.
- Tested CraftifAI's FirmGen tool to develop a complete product in the stipulated time and getting it featured on CraftifAI's official LinkedIn.

Technical Skills

Analog & PCB Design: LTspice, AFE Architecture, Bessel/Butterworth/KRC Filter Design, CMRR/Noise Optimization, KiCad - multi-layer, RF Layouts & mixed-signal PCB layout

Digital & RTL: VHDL, Verilog, RTL Synthesis, Xilinx Vivado, FPGA (Basys-3)

Embedded & Test Bench: Embedded C, STM32CubeIDE, ESP-IDF, I^2C/SPI/UART/RS-485 | DSO, Logic Analyzer, FFT Spectrum Analysis, SMT Rework